

Department of Information Engineering and Computer Science
KGE 2025 – Project Report

Knowledge Graph of Mongolian Cinema

Team-1

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Revision History

Version	Date	Description	Author(s)	Reviewer(s)
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Contributions

- Sainbuyan L.: Informal purpose, personas, scenarios, project report drafting, open issues, conclusion
- Munkhnaran B.: Concept identification and competency questions, Teleontology alignment, data definition and consolidation of data and ontology, dataset instantiation, entity,
- Altangerel N.: Relationships, ER conceptual model and focus classification, Teleontology conceptual model.
- Sainbuyan L. (Project Manager): Consolidation, editing and submission on behalf of the team.

PHASE 1-Purpose definition

1. Informal Purpose

When searching for information about Hollywood or Korean films, K-drama online, users can easily access comprehensive information in one place—release year, director, production company, genre, rating, viewer reviews, cast, and their related works. This is possible because these industries have established interconnected structured, knowledge ecosystems supported by mature knowledge graphs.

In contrast, the Mongolian film domain currently lacks such a unified, linked data ecosystem. Information about Mongolian cinema is scattered across many sources, unstructured, inconsistent and disconnected, difficult to integrate or analyze in a single search. Therefore, this project aims to:

- consolidate dispersed and unstructured Mongolian cinema data,
- build a connected data model using ontology and knowledge-graph methodology,
- provide a comprehensive, searchable, analyzable Knowledge Graph of Mongolian Cinema.

This knowledge graph will enable researchers, film enthusiasts, and media organizations to:

- explore connections among Mongolian films,
 - analyze relationships between filmmakers, actors, and studios,
 - study genres, award trends, and historical developments,
 - access all essential information about a film from a single query,
 - perform recommendations and graph-based analytics.
-

2. Personas

Persona 1: Non-Fan Viewer

Goal: Has no film knowledge. Wants to pick a high-rated film by a director/actor his girlfriend likes for a movie date.

Needs:

- View linked information among director, film, actors
- Runtime and short summary
- High-rated films featuring a preferred artist

Persona 2: Film Enthusiast

Goal: Find other works by a favorite actor or director.

Needs: Similar-genre films, related works by the same creators, ability to discover underrated, lesser-known but high-rated films. Importance placed on rating and release year.

Persona 3: Business Owner

Goal: Collaborate with filmmakers popular among target customers (e.g., children, women).

Needs: Networks of artist collaborations, film ratings and recent releases, identify current trending filmmakers with strong portfolios.

3. Scenarios

Scenario A: Non-Fan Viewer

Selecting a good Mongolian film for a date.

- **KG Needs:** Actor - list of other high-rated films; Movie - synopsis, runtime; Director – genre connections to understand film type.

Scenario B: Film Enthusiast

Finding underrated films by a favorite director.

- **KG Needs:** Director B – list of all directed films; Filter by high rating, low review count; Release year is before 2000.

Scenario C: Business owner

Collaborate with currently active, trending filmmakers.

- **KG Needs:** Production company - list of produced films; Filter by 2020–2024 high-rated releases; Identify frequently collaborating actors and directors.
-

4. Competency Questions (CQs)

The knowledge graph must be able to answer:

1. Can Mongolian films be listed by genre?
 2. What films has Director X made, and in which years?
 3. Which actors starred in Movie Y?
 4. Which Mongolian actors have the highest number of film appearances?
 5. Which Mongolian films have won domestic or international awards?
 6. What works has Production Company Z released?
 7. What common features link films of a specific genre?
 8. Can we generate an actor-actor or actor-director collaboration network?
 9. Who are the trending filmmakers of the last five years?
 10. Can we visualize the career timeline of an actor or director?
-

5. Concept Identification

Table 1: Concept Definition

Concept	Description
Movie	Metadata including release year, runtime, genre, rating, summary, awards, review co
Person	General category for creative individuals (actors, directors, producers, cinematograph
Actor	Subtype of Person. Performs roles in films.
Director	Subtype of Person. Oversees artistic and narrative aspects of a film.
Production Company	Entity responsible for financing, production, or distribution.
Genre	Category such as drama, comedy, action, arthouse, etc.
Rating	Numerical quality score (e.g., IMDB score, viewer rating).
Review Count	Number of user reviews or views (used for detecting underrated films).
Release Year	Year the film premiered.
Collaboration	Repeated or significant cooperation between artists (useful for Persona 3).

5.1. Core Concepts

6. Relationships

Table 2: Relationships

Relationship	From	To
Acted in	Actor	Movie
Directed by	Movie	Director
Produced by	Movie	Production Company
Has genre	Movie	Genre
Released in	Movie	Release Year
Has rating	Movie	Rating
Has reviews	Movie	Review Count

7. ER Diagram (Conceptual Model)

Entities:

- Movie: ID, Name, Release Year, Type, Genre, Rating, Runtime
- Actor: ID, Name, Birth date, Profession
- Director: ID, Name
- Production: ID, Name

Relationships:

- Movie → Actor (**acted_in**)
- Movie → Director (**directed_by**)
- Movie → Production (**produced_by**)

Table 3: Focus Classification

Concept	Focus Level
Movie	High
Actor	High
Director	High
Genre	Medium
Rating / Review Count	High
Production Company	Medium

8. Focus Classification

9. PF-Sheet

Aspect	Definition
Domain	Mongolian cinema (films, cast, crew, studios, genres, ratings)
Problem	Information is scattered, unstructured, and lacks interconnection
Objective	Create a connected, structured knowledge graph integrating all relevant film data
Scope	Mongolian films with metadata: title, release year, director, cast, production company, genre
Expected Users	Researchers, film enthusiasts, media organizations
Output	Knowledge graph (movies, actors, directors, production; edges: acted_in, directed_by, produced_by)

PHASE 2 - Information gathering

In this section, the second main input of the project is described, namely the list of data sources and datasets available for building the Knowledge Graph of Mongolian Cinema. The objective of this phase is to collect and organize the raw knowledge needed to build a unified, structured representation of Mongolian cinema data. Unlike Hollywood or Korean cinema ecosystems, which already benefit from mature, interconnected knowledge graphs that allow users to easily find release year, director, production company, genre, rating, reviews, and filmographies in one place, the Mongolian film domain lacks such a centralized knowledge infrastructure. Our goal is to gather, clean, standardize, and prepare these resources to enable the creation of such a connected ecosystem.

1. Sources Identification

- **Wikipedia Mongolia (mn.wikipedia.org)** Provides semi-structured pages for individual Mongolian films, including titles, directors, actors, production companies, runtimes, summaries, and award information. Type: Language + Data-value resource. Diversity Awareness: Not diversity-aware; structure varies between pages.
- **Kinosan.mn** A local Mongolian online movie portal listing films, posters, cast, genre tags, summaries, and sometimes ratings and release years. Type: Data-value resource. Diversity Awareness: Not diversity-aware; heterogeneous formats.
- **Domain Vocabulary Notes** Informal text resources describing film concepts (Movie, Person, Actor, Director, Genre, Rating). Type: Language resource. Diversity Awareness: Yes; domain vocabulary structures are fully under project control.

2. Datasets Collected

The following datasets were gathered and consolidated during the information gathering phase:

- **Movie Dataset** Contains metadata for Mongolian films:
 - Title, Release Year, Runtime
 - Genre
 - Rating
 - Actors
 - Director
 - Production Company
- **Actor Dataset** Includes:
 - Name
 - Last Name
 - Birthdate

- **Scraped Cast–Movie Mapping** Actor–Movie and Director–Movie relationships extracted from Wikipedia and Kinosan.mn. Used for constructing graph edges such as `acted_in` and `directed_by`.
- **Genre and Category Extraction** List of genres from scraped websites and manually curated classification rules.

3. Dataset Cleaning

A series of cleaning activities were required due to the heterogeneous and incomplete nature of Mongolian cinema sources.

- **Name normalization:** Actors and directors appear in multiple spellings on websites. Example: “ . ” vs “ - ”.
- **Duplicate removal:** Some films appear multiple times when scraped from different pages.
- **Missing values handling:** Several films lacked runtime, director, or review counts. Missing values were imputed as:
 - NULL values in the dataset
- **Standardization of release date formats:** From free-text (e.g., “2001 3- ”) to integer year values.

4. Dataset Standardisation

To support the creation of a unified Knowledge Graph schema, the datasets were standardized as follows:

- **Entity identifiers (IDs)** created for: Movies, Actors, Directors, Production Companies, and Genres.
- **Schema alignment:** All datasets aligned with a shared conceptual schema consistent with film ontologies and Schema.org patterns (`Movie`, `Person`, `Organization`, `CreativeWork`).
- **Relationship standardization:** Unified relationship labels:
 - `acted_in`
 - `directed_by`
 - `produced_by`
 - `has_genre`
 - `has_rating`
- **Attribute standardization:** Duration → integer (minutes) Release year → integer Rating → float Age rate → string

5. Discussion of Choices, Strong and Weak Points

Strengths

- **Local relevance:** The data set reflects the actual Mongolian cinema landscape, allowing realistic KG use cases.
- **Rich metadata coverage:** The combined sources provide: title, actors, directors, genre, release year, production company, ratings, summaries, and awards — enabling meaningful graph queries.
- **Compatibility with existing global models:** The data set can be mapped to Schema.org and Wikidata film structures to ensure interoperability.

Weaknesses

- **Heterogeneous formats:** Wikipedia and Kinolan pages differ significantly in structure and require extensive cleanup.
- **Incomplete data:** Many Mongolian films lack ratings, review counts, or award information.
- **Name inconsistency:** Lack of standardized transliteration or naming conventions for actors and directors.

Conclusion Despite limitations, the collected datasets (Wikipedia MN, Kinolan.mn) provide a sufficiently rich base to proceed to conceptual modeling, ontology design, and graph construction. The datasets have been cleaned, normalized, and standardized, making them ready for integration into the Mongolian Cinema Knowledge Graph.

PHASE 3 — Language Definition

Phase 3 of the iTelos methodology focuses on defining the formal representation language for the Mongolian Cinema Knowledge Graph. Based on the purpose definition and cleaned datasets from Phase 1 and Phase 2, this phase specifies the ontology classes, properties, and constraints used to represent cinematic knowledge in a structured and machine-readable form.

An ontology-based modeling approach is adopted to ensure consistency, extensibility, and compatibility with Semantic Web standards. The resulting language resources provide the basis for transforming tabular movie datasets into interconnected knowledge graph entities.

1. Ontology Scope and Design Principles

Domain: Mongolian Cinema

The ontology captures core information about Mongolian movies, including production entities and people involved in filmmaking, along with descriptive metadata.

Scope limitations:

- Focus on feature films produced in Mongolia.

- Only core production roles (Director, Actor) are modeled.
- Descriptive attributes are represented as literal values.

Design principles:

- Clear separation between entities (classes) and attributes (data properties).
 - Minimal but expressive modeling aligned with project competency questions.
 - Simplicity over excessive normalization.
-

2. Class Definitions

The ontology defines the following core classes:

- **Movie** — a cinematic work with descriptive and production metadata (core entity).
- **Production** — a production entity associated with a movie (e.g., production company or university).
- **Director** — a person who directs a movie.
- **Actor** — a person who performs in a movie.

These classes represent the primary real-world entities required to answer the defined competency questions.

3. Object Properties

Object properties describe relationships between classes.

Table 4: Object Properties

Property	Domain	Range
hasProduction	Movie	Production
hasDirector	Movie	Director
hasActor	Movie	Actor

These properties connect movies to the people and organizations involved in their creation.

4. Data Properties

Data properties capture literal attributes associated with entities.

Movie

- genre (`xsd:string`) — genre of the movie
- runtime (`xsd:time`) — duration of the movie
- releaseYear (`xsd:gYear`) — year of release
- rate (`xsd:string`) — short descriptive rating or review

Person and Production

- name (`xsd:string`) — name of a person or production entity
-

5. Modeling Constraints and Assumptions

The following constraints are applied to simplify modeling and ensure consistency:

- Each Movie is linked to at most one Director.
- A Movie may be linked to multiple Actors.
- A Movie may be linked to one Production entity.
- All descriptive values are modeled as literals.

These assumptions reflect the structure of the available datasets and project scope.

6. Dataset-to-Ontology Mapping

Fields from the Phase 2 datasets are mapped to ontology elements as follows:

Table 5: Dataset to Ontology Mapping

Dataset Field	Ontology Element
Movie title	Movie
Production name	Production.name + hasProduction
Director name	Director.name + hasDirector
Actor name	Actor.name + hasActor
Genre	Movie.genre
Runtime	Movie.runtime
Release year	Movie.releaseYear
Rate / description	Movie.rate

This mapping enables systematic conversion of tabular data into RDF-style triples.

7. Example Knowledge Graph Instance

The following example illustrates an instantiated movie entity:

Movie:

```
genre: Drama
runtime: 01:45:00
releaseYear: 1995
rate: Highly regarded classic
hasDirector: Director_ .
hasActor: Actor_1, Actor_2
hasProduction: Production_Entity
```

This example demonstrates how real-world movie data is represented using the defined language resources.

PHASE 4 - Knowledge definition

This section describes the Knowledge Definition phase (Phase 4) of the project. Following the structure of the previous phases, it details the sub-activities performed by all team members, the methodological choices made, and the outcomes produced. The objective of this phase is to transform the ontology and aligned datasets into a concrete, goal-oriented Knowledge Graph of Mongolian Cinema.

Knowledge Definition sub-activities:

- **KTelos**
 - **Teleology definition**
 - **Teleontology definition**
- **Dataset cleaning and formatting**

1. Teleology Definition

The teleology defines the purpose and intended use of the knowledge to be represented. In this project, the primary goal is to enable structured exploration, analysis, and discovery of Mongolian cinema data, which is currently fragmented and poorly interconnected.

The defined teleological objectives are:

- enable users to discover Mongolian films by genre, rating, release year, and creators,
- support analysis of actor–director and artist collaboration networks,
- allow identification of trending filmmakers and production companies,
- provide a unified and searchable representation of film-related knowledge.

These goals directly correspond to the competency questions and user scenarios defined in Phase 1.

2. Teleontology Definition

The teleontology links the defined goals to concrete ontology elements. Each class, property, and relationship in the ontology is justified by its contribution to fulfilling at least one teleological objective.

Core ontology concepts such as *Movie*, *Actor*, *Director*, and *ProductionCompany* support exploration and discovery tasks, while relationships such as *acted_in*, *directed_by*, and *produced_by* enable network analysis and collaboration-based queries. Data properties such as rating, review count, and release year allow filtering, ranking, and trend detection.

This teleontology ensures that the Knowledge Graph is not only structurally correct but also purpose-driven and aligned with user needs.

3. Dataset Cleaning and Formatting

To enable reliable knowledge instantiation, datasets collected in Phase 2 were further cleaned and formatted during Phase 4. This included:

- resolving remaining name inconsistencies across data sources,
- ensuring consistent use of identifiers for all entities,
- validating data types for numerical and textual attributes,
- formatting datasets to match ontology class and property definitions.

These steps ensured seamless transformation from tabular data into graph-based knowledge representations.

4. Choices, Strengths, and Weaknesses

Several design choices were made during the Knowledge Definition phase. A key strength of the adopted approach is its clear alignment between user goals, ontology structure, and datasets, resulting in a coherent and reusable Knowledge Graph. The use of teleontology ensures that all modeled knowledge serves a well-defined purpose.

However, some limitations remain. Certain datasets contain incomplete or missing information, which restricts the depth of analysis for some films or artists. Additionally, simplifying assumptions, such as assigning a single director per movie, reduce modeling complexity but may limit expressiveness in future extensions.

Despite these limitations, the outcomes of Phase 4 provide a solid and extensible foundation for knowledge graph instantiation and future analytical or recommendation-oriented applications.

PHASE 5 - Entity definition

This section describes the Entity Definition phase (Phase 5) of the iTelos methodology. The objective of this phase is to integrate the knowledge structures defined in the Knowledge Definition phase with the cleaned and standardized datasets, producing the final, concrete Knowledge Graph (KG) of Mongolian Cinema.

Unlike previous phases, the Entity Definition phase operates on a unified layer where knowledge and data are merged. The activities carried out in this phase focus on entity recognition and matching, followed by entity-to-ontology mapping, supported by the Karma data integration tool.

5.1 Input Resources

The input resources for this phase include:

- Cleaned and normalized datasets produced during Phase 2 and Phase 4, including Movies, Actors, Directors, and Production Companies.
- The ontology and teleontology defined in the Knowledge Definition phase.
- The formalized user purpose and competency questions defined in Phase 1.

All datasets were organized into a single Excel workbook composed of multiple sheets, each corresponding to a specific entity type or relationship. This structure was designed to simplify the mapping process within Karma.

5.2 Entity Recognition and Matching

Entity recognition and matching aim at identifying real-world entities and resolving different representations of the same entity across datasets. This activity was particularly critical for actor entities, whose names often appear in abbreviated or heterogeneous forms in movie cast descriptions.

A normalization-based matching strategy was adopted, consisting of:

- Text normalization (lowercasing, punctuation removal, whitespace normalization).
- Removal of noise expressions such as “ ” or “ ”.
- Matching of normalized actor names against the cleaned Actor dataset.
- Assignment of unique identifiers when a match was identified.

Entities that could not be automatically matched were retained as unmatched instances and explicitly documented, ensuring transparency and enabling future refinement.

5.3 Entity Mapping with Karma

Entity mapping was performed using the Karma tool. Each dataset sheet was mapped to the corresponding ontology class, while dataset columns were aligned with ontology properties.

The main mappings defined are summarized as follows:

- **Movies** → *Movie*
- **Actors** → *Actor*
- **Directors** → *Director*

- **ProductionCompanies** → *ProductionCompany*

Relationship datasets were mapped to ontology object properties:

- MovieActor_edges → **acted_in**
- MovieDirector_edges → **directed_by**
- MovieCompany_edges → **produced_by**

Karma’s visual mapping interface ensured consistency between the data layer and the teleontology defined in the previous phase.

5.4 Final Knowledge Graph Generation

Once entity recognition and mapping activities were completed, Karma was used to generate the final Knowledge Graph. The graph was exported in RDF format using Turtle serialization.

The resulting Knowledge Graph integrates movies, actors, directors, production companies, and their relationships into a unified, machine-readable structure. The final KG satisfies the project purpose and supports the competency questions defined at the beginning of the project.

5.5 Strengths and Limitations

Strengths

- Strong alignment between user purpose, ontology, and data.
- Explicit handling of entity ambiguity and incomplete information.
- Reusable and extensible Knowledge Graph structure.

Limitations

- Some entities could not be fully matched due to inconsistent name representations.
- Missing metadata for certain films limits the depth of analysis.

Open issues and conclusion

While the project establishes a solid foundation for a Mongolian film knowledge graph, addressing data heterogeneity, incompleteness, and naming inconsistencies remains essential for advanced analytics and comprehensive cultural insights.

- Heterogeneous data formats: Wikipedia and Kinosan.mn differ significantly in structure, requiring extensive data cleaning.
- Incomplete data: Many Mongolian films lack ratings, review counts, or award information.
- Name inconsistency: No standardized transliteration or naming conventions for actors and directors.
- Limited coverage: Older and lesser-known films are not fully represented.

Future work and opportunity:

- Expand data sources to include archives, awards, and review platforms.
- Improve name normalization and transliteration strategies.
- Extend the ontology to support multiple directors and richer relationships.
- Explore connections among films, filmmakers, actors, and studios.
- Analyze genres, award trends, and historical developments.
- Enable single-query access to all essential film information.